**INFOSYS SPRINGBOARD ONLINE INTERNSHIP**

HotelRevAI

AI Driven Revenue Analysis for Hotels

**Milestone 3-Submission**

Data Collection and Preparation,DashBoard Implementation and Business Metrics Phase

Submitted By

A Veda Varshini

Computer Science and Engineering (Data Science)

Vellore Institute Of Technology (VIT), Vellore

Mentor Name: Selciya

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# 1. Abstract

The hospitality industry generates large amounts of booking data, but analyzing this data effectively is challenging due to missing values, inconsistent records, and irrelevant information. Without proper data preparation, hotels struggle to understand revenue trends and customer booking behavior.

This project focuses on analyzing hotel booking data to improve revenue management using data analytics tools. The dataset used in this project was collected from Kaggle and includes booking details such as stay duration, cancellation status, customer information, and average daily rate. The work completed so far includes dataset collection, cleaning, and transformation using Power BI and Power Query.

Several preprocessing operations were performed, including removal of unnecessary and sensitive columns, handling missing values, creation of new analytical columns such as total nights, total guests, arrival date, and revenue per booking, and removal of invalid or cancelled bookings. The cleaned dataset will be used in later stages to develop dashboards and revenue insights to support data-driven decision-making in hotel management.

# 2. Introduction

Hotels depend on efficient revenue management strategies to maximize profits and maintain occupancy. However, raw booking data often contains inconsistencies and missing values, making analysis difficult.

This project aims to transform raw hotel booking data into a structured dataset for revenue analysis. By using Power BI tools, hotels can analyze booking trends, understand customer behavior, and improve revenue strategies. The proposed solution involves cleaning

and preprocessing booking data and then creating analytical dashboards for better decision-making.

## 3. Problem Statement

Hotels face challenges in understanding revenue fluctuations, booking patterns, and customer segments due to large volumes of raw booking data. Without proper visualization and analysis, it becomes difficult to identify peak seasons, high-revenue sources, and the impact of cancellations. This project aims to solve this problem by using Power BI to analyze hotel booking data and create interactive dashboards for effective business insights.

## 4. Objectives of the Project

- Analyse overall revenue performance and identify growth trends over time
- Analyse hotel performance using ADR, cancellation rate, total booking nights, total guests.
- Analyse customer behaviour booking patterns by customer type, room type, repeated customers.
- Create interactive dashboards using Power BI that includes filtering by year, customer behaviour charts, booking patterns that provide meaningful visual insights.
- By all these to support data driven decision making that is useful to business.

## 5. Dataset Overview

This is hotel booking dataset collected from Kaggle. This data set contains booking information for a city hotel and a resort hotel, and includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things.

## 6. Data loading and cleaning

- The dataset was imported into Power BI from a CSV file containing hotel booking records. Power BI's Power Query Editor was used to preview, transform, and prepare the data before loading it into the data model.

• Data preprocessing was performed using Power Query to ensure accuracy and consistency. The following general cleaning operations were carried out:

- Handled missing or null values in important columns.
- Verified and corrected data types (date, numeric, text).
- Removed duplicate records.
- Removed columns that are not participating in

revenue analysis • Created new calculated columns such as:

- Total Stay Duration (weekend nights + weekday nights) ○ Total guests (adults + children + babies)
- Revenue-related measures using ADR and stay duration. ○ Converted arrival date columns into single column.

## 7.Data Modelling

Data modelling is a crucial step in Power BI that defines the logical relationships between different tables and ensures efficient filtering and aggregation across visualizations. In this project, a structured relational data model was developed to support multi-dimensional analysis of hotel booking and revenue data.

The data model consists of the following primary

components:

• **Fact Table (Hotel_Data):**

This table contains transactional booking-level data such as customer type, ADR, reserved room type, booking changes, country, and arrival date.

• **Dimension Table (Calendar):**

This table includes:

- Date

- Month
- Month Number
- Year

- **Measures Table (d3_measures):**

  A dedicated measures table was created to store calculated metrics using DAX (Data Analysis Expressions). These measures include:

### Total Revenue

```
Total Revenue =
SUM(Hotel_data[Revenue])
```

### Total Bookings

```
Total Bookings =
COUNTROWS(Hotel_data)
```

### Total Customers

```
Total Customers =
COUNTROWS(Hotel_data)
```

### Total Guests Count

```
Total Guests Count =
SUM(Hotel_data[Total Guests])
```

### Total Nights Stayed

```
Total Nights Stayed =
SUM(Hotel_data[Total Nights])
```

### Revenue per Night

```
Revenue per Night =
DIVIDE(
 [Total Revenue],
 [Total Nights Stayed],
 0
)
```

### Rev_per_Night

```
Rev_per_Night =
```

```
DIVIDE(
 SUM(Hotel_data[Reven
ue]),  SUM(Hotel_data[
Total Nights]),  0
)
```

## Cancelled Bookings

```
Cancelled Bookings =
COUNTROWS(
 FILTER(
 Hotel_data,
 Hotel_data[is_cancele
d] = 1  )
) + 0
```

## Cancellation Rate

```
Cancellation Rate =
DIVIDE(
 [Cancelled Bookings],
 [Total Bookings],
 0
)
```

## Revenue Loss

```
Revenue Loss =
CALCULATE(
 SUM(Hotel_data[Re
venue]),  Hotel_data[
is_canceled] = 1 )
```

## Cancellation Impact

```
Cancellation Impact =
DIVIDE(
 [Revenue Loss],
 [Total Revenue],
 0
)
```

## Bookings_New

```
Bookings_New =
```

```
CALCULATE(
 COUNTROWS(Hotel_da
ta),  Hotel_data[is_repeate
d_guest] = 0 )
```

## Bookings_Repeated

```
Bookings_Repeated =
CALCULATE(
 COUNTROWS(Hotel_da
ta),  Hotel_data[is_repeate
d_guest] = 1 )
```

## Repeated Guests

```
Repeated Guests =
CALCULATE(
 SUM(Hotel_data[Total
Guests]),  Hotel_data[is_r
epeated_guest] = 1 )
```

## Repeat %

```
Repeat % =
DIVIDE(
 [Bookings_Repeated],
 [Total Bookings],
 0
)
```

## Occupancy Proxy

```
Occupancy_Proxy =
DIVIDE(
 [Total Guests Count],
 [Total Nights Stayed],
 0
)
```

## Revenue_Repeated

```
Revenue_Repeated =
CALCULATE(
 SUM(Hotel_data[Revenue]),
 Hotel_data[is_repeated_guest] = 1
```

)

The above DAX queries are used to create calculated columns, measures, and calculated tables in Power BI.

A **one-to-many relationship** was established between:

Calendar[Date] → Hotel_Data[Arrival Date]

This relationship allows accurate time-based filtering and aggregation across dashboards.

## 8.Dashboard Implementation

Three interactive dashboards were developed using Power BI to present insights related to hotel booking performance, customer behaviour, and revenue optimization.

**Dashboard 1: Hotel Revenue Performance Dashboard**

This dashboard provides a consolidated overview of overall hotel performance through key performance indicators (KPI) such as **Total Revenue, Total Bookings, Average Daily Rate (ADR), and Total Guest Count.** Additionally, it includes visualizations that represent:

- Monthly Revenue Trends
- Booking Distribution by Market Segment
- Distribution Channel Analysis
- Room Type Demand
- Revenue Trend Over Time
- Customer Type Booking Distribution

These visualizations assist in identifying seasonal demand patterns and revenue-generating customer segments.

The integrated analysis reveals:

- Strong reliance on online booking channels.
- Seasonal demand patterns affecting revenue.
- High contribution from transient customers.
- Concentrated demand for specific room types.

- Stable pricing strategy reflected in ADR.

**Dashboard 2: Customer and Booking Behaviour Dashboard**

This dashboard focuses on analysing customer booking behaviour and engagement patterns. It includes KPI cards such as **Total customers, Average lead lite, Total repeated customers, Repeated guests percentage**. It provides insights into:

- Booking Distribution by Customer Type
- Top 10 Booking Countries
- Revenue Contribution from Repeated vs New Guests
- Booking Trends based on Lead Time Group

The analysis helps in understanding customer loyalty patterns and booking preferences across different regions.

**The dashboard reveals:**

- Strong customer inflow.
- Low repeat guest ratio.
- High early booking behavior.
- Demand concentrated in selected countries.
- Dominance of transient travelers.

**Dashboard 3: Revenue Optimization and AI Insights Dashboard**

This dashboard focuses on Revenue Optimization & AI Insights. It includes KPI cards such as **Revenue per Night, Cancellation Impact, Occupancy Proxy, Estimated Revenue Loss.** It provides insights into:

- Total Revenue by Year
- Average ADR by Year
- Revenue Variation with Lead Time
- Demand Distribution by Room Type

These insights support strategic decision-making related to pricing policies and booking management.

**The Revenue Optimization dashboard reveals:**

- Stable nightly revenue performance

- Increasing pricing strategy effectiveness
- Strong early booking contribution
- Efficient room utilization
- Balanced revenue growth trend

## 9. Business Metrics

### 1 Revenue Metrics

- **Total Revenue** – Overall income generated
- **Revenue per Night** – Average income per occupied night
- **Revenue Trend (Monthly/Yearly)** – Seasonal & growth analysis
- **Revenue Loss** – Income lost due to cancellations
- **Cancellation Impact (%)** – Financial impact of cancellations

### 2 Booking & Demand Metrics

- **Total Bookings** – Total reservations
- **Average Lead Time** – Avg. days between booking & arrival
- **Revenue by Lead Time Group** – Most profitable booking window
- **Demand Index** – Monthly booking intensity

### 3 Customer Metrics

- **Total Customers** – Total booking customers
- **Repeated Customers** – Returning guests
- **Repeat Guest %** – Customer retention rate
- **Bookings by Customer Type** – Transient, Contract, Group

### 4 Occupancy Metrics

- **Occupancy Rate / Proxy** – Room utilization level
- **Weekend Ratio (%)** – Share of weekend bookings

### 5 Market & Channel Metrics

- **Revenue by Market Segment** – Online TA, Direct, Corporate
- **Revenue by Distribution Channel** – TA/TO, Direct, GDS
- **Bookings by Country** – Top demand locations

### 6 Pricing & Room Metrics

- **Average Daily Rate (ADR)** – Avg. room price

- **ADR Trend** – Pricing evolution over time
- **Demand by Room Type** – Most preferred rooms

These business metrics provide insights into revenue performance, customer behavior, demand patterns, pricing strategy, and occupancy efficiency to support revenue optimization decisions.

## 10.Conclusion

The Hotel Revenue AI project successfully transformed raw hotel booking data into meaningful business insights through data cleaning, modeling, and interactive dashboard development.

By implementing structured data modeling and DAX-based calculations, the system enables comprehensive analysis of revenue performance, customer behavior, occupancy trends, and pricing strategies.

The dashboards provide real-time insights into booking patterns, seasonal demand fluctuations, lead-time behavior, and room-type performance. These insights support informed decision-making and help optimize revenue management strategies.

Overall, the project demonstrates how data analytics and AI-driven insights can enhance operational efficiency, improve customer targeting, and maximize hotel profitability.